

Numerical Methods — Laboratory Activity 02

Reservoir Stage Curve Fitting — Results

Model: Four-parameter logistic function $h(t) = c + a/(1 + \exp(-k(t - t_{\text{max}})))$

Initial guess: $p_0 = [14.10, 0.50, 0.12, 11.0]$ Readings: 96 $\Delta t = 0.25$ h $df = 92$

Parameter	Estimate	Std. Error	t	Two-tailed p
c	14.10600579	0.05017039	281.161946	<0.000001
a	0.48529977	0.08958716	5.417069	0.000000482
k	0.11998783	0.02650435	4.527099	0.000017867
t_{max}	10.99007434	0.89855038	12.230894	<0.000001

Goodness-of-fit / residual results	Value
SSE	0.0185999466
SST	0.8244906250
R ²	0.9774406815
Standard error, s	0.0142187669 m
Maximum residual	0.035161 m
Time of maximum residual	10.75 h

Area / derivative result	Value
Maximum central dh/dt	0.100000 m/h
Time of maximum central dh/dt	1.50 h (01:30)
Area by scipy.integrate.quad	341.0433284462 m·h
quad absolute error estimate	3.7863×10^{-12}
Raw-data trapezoid	341.0475000000 m·h
Difference	0.0041715538 m·h

Residual reading: The residuals are generally centered around zero, but short runs of positive and negative residuals remain. The logistic model captures the overall reservoir rise but does not reproduce every short-term centimeter-level fluctuation. The area is in meter-hours (m·h), not volume.